



COURSE SYLLABUS						
Cours e Code	COMP 001	Course Title	Introduction to Computing		Course Credit	3
Course Description		This course is designed to provide students with a breadth-first overview of computing fundamentals. The materials covered in this course include concepts such as components of the computer system, number system operation and conversion, data representation, digital logic system, levels of programming, computer networks, computer applications, current trends and issues.				
Pre-Requisites		None	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
		Apply knowledge of computing, science, and mathematics appropriate to the discipline.								



	Analyze complex problems, and identify and define the computing requirements appropriate to its solution												
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Learn the current issues surrounding the development and use of ICTs								
	Evaluate the positive and negative side of a critical IT-related issues								



	Learn the risks involved in using ICT, as well as, the protection and security measures to avoid becoming victims of cybercrimes								
	Identify emerging and converging technologies, including trends in hardware, software and data security								
	Develop cooperation in group discussion, research activities and presentations								
	Present the Pros and Cons of ICT's societal impact								
	Discuss the important role of ICTs in the improvement of the modern society, as well as its effects in the evolving the cultureof things								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignmen t to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	Introduction to the Course a. Vision Mission Goals and Objective of the University, and College. b. Self-Introducti on c. Course Overview d. Grading System Classroom Management	Vision Mission Goals and Objective of the University, and College. Self-Introduction Course Overview Grading System Classroom Management	Demonstrate a sense of readiness for the upcoming semester. Identify their learning outcomes and expectations for the course. Recognize their capacity to create new understandings from reflecting on the course	Orientation Self-Introduction Group Discussions			X			
2	Unit 1: Overview of Information and Communications Technology	a. Categorize computers. b. Contrast elements of computer system. c. Identify various events/improvements in the computing world. d. Qualify the understanding of computer usage	Lectures/Discussions/Video Presentation	Lectures/Discussions/Video Presentation	Activity, Short Quiz					



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st
POLYTECHNICU





	1. Introduction to Computers 2. Elements of a Computer System 3. Classification of Computers 4. Capabilities and Limitations of Computers 5. History of Computing								
3-8	Unit 2: Data Representation A. Number System Operation and Conversion B. Data Representation 1. Numeric a. Unsigned b. Signed c. Fixed d. Floating Non-numeric	a. Distinguish the various number system and data representation b. Perform number system operation such addition, subtraction and complement. c. Explain the workflow of the processor/memory/input and output devices and architecture d. Perform number conversion, fixed point and floating point number representation.	Lectures/Discussions Video Presentation	Lectures/Discussions Video Presentation	Long Quiz				
9	MIDTERM EXAMINATION								
10-11	Unit 3: Hardware 1. Digital Logic System 2. Processor 3. Memory	a. Explain the functions of the main units of a physical computer system b. Describe what input and output devices are c. Differentiate the primary and secondary storage devices	Lectures/Discussions/Video Presentation	Lectures/Discussions/Video Presentation	Short Quiz				



	4. Input and Output Devices	d. Give examples of I-O devices e. Define what boolean algebra is f. Identify the different logic gates g. Illustrate the representation of the different logic gates h. Convert boolean algebra expression into a logic circuit i. Create truth tables for the corresponding logic circuits and boolean expression j. Explain basic theorems and postulates on digital logic system							
12	Unit 4: Peopleware 1. Roles and Job Titles 2. Code of Ethics for ICT Professionals	a. Contrast roles and jobs in the ICT profession. b. Summarize and report insights gained from ICT professionals. c. Discuss a typical day of an ICT professional.	Lectures/Discussions/Video Presentation	Lectures/Discussions/Video Presentation	Short Quiz				



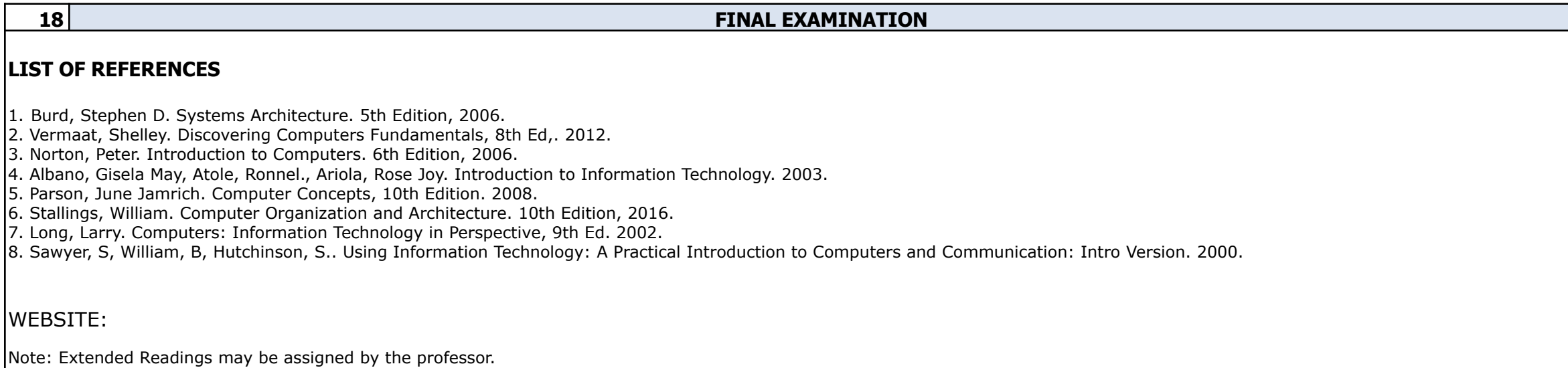
12	Unit 5: Software 1. Assembly and Machine Language 2. Compilers and Translators 3. Programming Languages 4. Operating Systems 5. Application Software	a. Differentiate an application software from a system software b. Explain the types of a system software c. Explain the different functions of an operating system d. Discuss the different means by which application softwares are made available	Lectures/Discussions/Video Presentation	Lectures/Discussions/Video Presentation	Short Quiz					
13	Unit 6: Computer Networks 1. Concepts in Computer Networking 2. Network Services 3. Internet and the World Wide Web	a. Discuss network models b. Explain network topologies c. Explain internet concepts	Lectures/Discussions/Video Presentation	Lectures/Discussions/Video Presentation	Short Quiz					
14	Unit 7: Computer Trends and Issues in ICT 1. Current Trends 2. Social and Legal Issues	a. Explains the current ICT trends and social issues. b. Initiates discipline and relates knowledge of ICT trends and issues on study works.	Lectures/Discussions/Video Presentation	Lectures/Discussions/Video Presentation	Short Quiz					



15-16	Unit 8: Special Interest Topics in ICT 1. Artificial Intelligence 2. Data Science 3. Social Networking and Society	a. Explain the difference between AI, machine learning, and deep learning b. Provide applications of AI in different industries and in daily use. c. Identify important milestones in the history of AI d. Explain supervised, unsupervised learning and other concepts related to AI e. Explain what the field of data science is f. Identify the skills/expertise needed to be a data scientist g. Discuss what big data is and how it relates to data science h. Discuss where each specific popular media sites are commonly used i. Analyze the benefits of social media to society j. Discuss the disadvantages of social media	Lectures/Discussions Video Presentation Group Discussion and Presentation	Lectures/Discussions Video Presentation Group Discussion and Presentation	Short Quiz					
17	FINAL EXAMINATION									



18	Round-up Activities	a. Differentiate Patents and Copyrights b. Identify and Enumerate the different lincenses used by FOSS c. Explain how IPR and licenses directly changes the course of technological development	Lectures/Discussions Video Presentation Group Discussion and Presentation	Lectures/Discussions Video Presentation Group Discussion and Presentation	Short Quiz					
----	---------------------	---	---	---	------------	--	--	--	--	--



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: - Take time to read and watch the provided class learning materials. - Fulfill class requirements such as posting on discussion boards and submitting group presentations. - Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.

THE COUNTRY'S 1st
POLYTECHNICU





POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st
POLYTECHNICU





POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st
POLYTECHNICU





POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st
POLYTECHNICU





The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.

Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st
POLYTECHNICU



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
--	---

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA

THE COUNTRY'S 1st
POLYTECHNICU

**WORLD UNIVERSITY RANKINGS**

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

**WURI**
The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/PRODUCTION

**QS STARS**

Employability
Online Learning

Teaching
Academic Development
Inclusiveness

Social Responsibility
Specialist Online Bachelor of Science in Nutrition and Dietetics (SOND)

**Times Higher Education Impact Rankings 2024**
DEBUTS AT 1501+



Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Quizzes		Quizzes	
Long Test		Long Test	
Recitation and Attendance		Recitation and Attendance	
Research Work		Research	
Assignment	30%	Work	
Projects		Assignment	30%
Exercises		Projects	100%
Midterm Examinations		Exercises	
	100%	Final Examinations (Teaching Demonstration)	

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.
COURSE REQUIREMENT/S	GRADING SYSTEM



- Making a Learning Plan in Teaching Prose (focusing on Completeness, Appropriate Approach Used, and Facilitative Process) - Teaching Demonstration in Prose or poetry (focusing on Teacher- Like Simulation, Preparation, Classroom Management, Learning Environment, and Interest and Diligence in Teaching Poetry)	FINAL GRADE = (FIRST GRADING + SECOND GRADING) / 2		
CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section:		Name of Faculty: MODESTO, LADY MELINDA MINETTE L.	
Day and Time:		Consultation Time:	
Room:		Office Tel. No./ Mobile Phone No.	
School Year:	Semester:	E-Mail Address:	



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
PROF. LADY L. MODESTO Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA

THE COUNTRY'S 1st
POLYTECHNICU

**WORLD UNIVERSITY RANKINGS**

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

**WURI**
The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI
CAPABILITY

Rank 39
STUDENT SUPPORT
CAPABILITY

Rank 50
SYMBOL/
CAPABILITY

**QS STARS**

**5**
Employability
Online Learning

**4**
Teaching
Academic Development
Inclusiveness

**3**
Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

**Times Higher Education**
Impact Rankings 2024
DEBUTS AT 1501+